

BAHLAOUI AHMED

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SUMMARY

AI & Machine Learning engineering student with strong foundations in statistics, machine learning, and scientific computing. Experienced in Python-based data analysis, preprocessing large datasets, feature engineering, and building reproducible ML workflows using pandas, scikit-learn, TensorFlow, and PyTorch.

QUALIFICATIONS

Ecole Nationale des Arts et Métiers

Rabat, Morocco

Engineering Degree, AI & Machine Learning

Sep 2023 – Jun 2028

- Coursework: Machine Learning, Probability & Statistics, Algorithms, Data Structures, Mathematical Optimization, Deep Learning, Natural Language Processing, Computer Vision, Data Science, Software Engineering

PROJECT EXPERIENCE

GhostDrop | TypeScript, Fastify, Svelte, Redis, PostgreSQL, MinIO, Docker | 2026

- Built an end-to-end encrypted (AES-256-GCM), anonymous file-sharing platform with ephemeral transfer sessions, inspired by Wormhole and Send Anywhere
- Designed a streaming-first upload/download pipeline to MinIO object storage with Redis-backed rate limiting and TTL-based session cleanup
- Unified Caddy gateway serving the frontend and reverse-proxying the API over HTTP/3 + QUIC, with Zod-validated APIs and an auto-migrating Docker staging environment

Face-cluster | Python, ONNX Runtime, Gradio | 2026

- Built an unsupervised face clustering pipeline grouping unlabeled images by identity using ONNX-accelerated face embeddings and density-based clustering
- Accelerated inference with GPU-backed ONNX Runtime and exposed the pipeline through an interactive Gradio interface

Moroccan Used Car Price Prediction | Python, Pandas, scikit-learn, XGBoost | 2026

- Built an end-to-end ML pipeline on 100K+ real-world records, cleaning 25K+ missing/placeholder values and mixed Arabic/French categorical data
- Engineered features from mileage intervals, fiscal power, and brand/model/geographic variables; compared models via RMSE, MAE, and cross-validated R^2
- Reached $R^2 = 0.91$ with XGBoost and deployed an interactive prediction app on Streamlit Cloud

Task Management API (in progress) | Java, Spring Boot, PostgreSQL | 2026

- Building a REST API with Spring Security (JWT auth), role-based access control, and JPA/PostgreSQL persistence to apply Spring Boot fundamentals in a real backend context

TECHNICAL SKILLS

- **Programming:** Python, Java, C++, TypeScript, SQL
- **Backend & Deployment:** Spring Boot, FastAPI, Node.js, REST APIs, Docker, Linux, Git, MLflow
- **Data Science:** Pandas, NumPy, Matplotlib, Seaborn, Jupyter
- **Machine Learning:** scikit-learn, TensorFlow, PyTorch, XGBoost, ONNX Runtime, Feature Engineering
- **Databases:** PostgreSQL, MongoDB, Redis, MySQL
- **Scientific Computing:** Probability, Statistics, Numerical Methods, Optimization

CERTIFICATIONS

- Data Engineering Track — DataCamp, 2026
- Supervised Machine Learning: Regression and Classification — Coursera, 2025
- Statistics Foundations — Meta, 2024

LANGUAGES

- Arabic (Native), French (Fluent), English (Fluent)